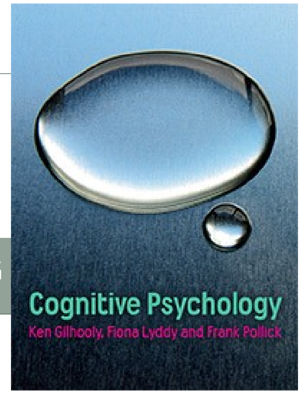
CHAPTER 3

REASONING & DECISION MAKING

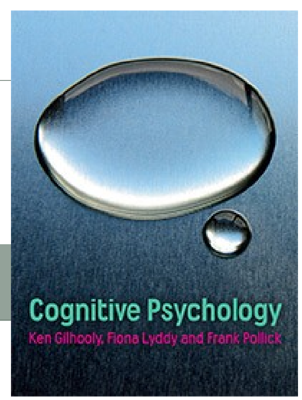


TWO MAIN SECTIONS

- 1. Deduction
- 2. Decision Making

1. DEDUCTION

INTRODUCTION



WHAT IS DEDUCTIVE REASONING?

<u>Problem 1:</u>
All A's are B's.
All B's are C's.
What follows?

WHAT IS DEDUCTIVE REASONING?

Problem 1:
All A's are B's.
All B's are C's.
What follows?
All A's are C's.

WHAT IS DEDUCTIVE REASONING?

Problem 1:
All A's are B's.
All B's are C's.
What follows?
All A's are C's.

Problem 2:
None of the A's are B's.
All B's are C's.
What follows?

WHAT IS DEDUCTIVE REASONING?

Problem 1:
All A's are B's.
All B's are C's.
What follows?
All A's are C's.

Problem 2:
None of the A's are B's.
All B's are C's.
What follows?
Some of the C's are not A's.

WHAT IS DEDUCTIVE REASONING?

Problem 1:
All A's are B's.
All B's are C's.
What follows?
All A's are C's.

Problem 2:
None of the A's are B's.
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Some of the C's are not A's.

- All info for conclusion in Problem

WHAT IS DEDUCTIVE REASONING?

Problem 1:
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What follows?
All A's are C's.

Problem 2:
None of the A's are B's.
All B's are C's.
What follows?
Some of the C's are not A's.

- All info for conclusion in Problem
- Some problems are easy, some difficult

DIFFERENCE WITH INDUCTION

- *Swan 1 is white*
- *Swan 2 is white*
- *…*
- *Swan 100 is white*
- *So all swans are white.*
- You go further than the info in premises

DIFFERENCE WITH INDUCTION

- *Swan 1 is white*
- *Swan 2 is white*
- *…*
- *Swan 100 is white*
- *So all swans are white.*

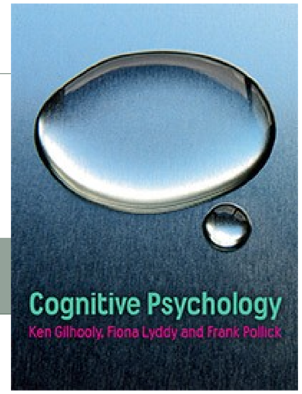
POTENTIAL CRITICISM

- Why deductive reasoning, because **not relevant** for daily life?

- All syllogisms are problems which are irrelevant for daily life.
- All problems which are irrelevant for daily life should not be studied by cognitive scientists.
- Therefore, all syllogisms should not be studied by cognitive scientists.

1. DEDUCTION

MENTAL LOGIC



MENTAL LOGIC

- Translation into abstract structure (p's, q's, …)
- + syntactic rules
- But: not all rules of logic
  - We make mistakes
- Example:
- modus ponens

  if p then q
  p
  ∴ q
  MP-rule is part of rule-set

MENTAL LOGIC

- Translation into abstract structure (p's, q's, …)
- + syntactic rules
- But: not all rules of logic
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- Example:

modus ponens	modus tollens
if p then q	*if p then q*
p	*not-q*
∴ q	∴
MP-rule is part of rule-set	

MENTAL LOGIC

- Translation into abstract structure (p's, q's, …)
- + syntactic rules
- But: not all rules of logic
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- Example:

modus ponens	modus tollens
if p then q	*if p then q*
p	*not-q*
∴ q	*∴ not-p*
MP-rule is part of rule-set	MT-rule not

HOW MODUS TOLLENS? INDIRECT!

1. *if p then q*
2. *not-q*
3. *p* (hypothesis)
4. *q* (MP of 1 and 3)
5. *q and not-q* (conjunction of 4 and 2)
6. *not-p* (reductio ad absurdum)

- Two important principles:
  - Number of steps
  - Availability of rules
- Matter of empirical research

THE WASON SELECTION TASK

Here are 4 cards. Each of them has a letter on one side and a number on the other side.

Two of these cards are shown with the letter side up, and two with the number side up.

E	C	5	4

Indicate which of these cards you have to turn over in order to determine whether the following claim is true: **If a card has a vowel on one side, then it has an odd number on the other side.**

THE BEER/COKE TASK

In its cracking down against drunk drivers, New York law enforcement officials are revoking liquor licences left and right. You are a bouncer in a Boston Bar, and you'll lose your job unless you enforce the following law:

If a person is drinking beer, then he must be over 20 years old.

The cards below have info about 4 people sitting at a table in your bar. Each card represents one person. One side of the card tells what a person is drinking, the other side the person's age. Indicate only those card(s) your definitely need to turn over to see if any of these people are breaking the law.

Drinking beer	Drinking Coke	25 years old	16 years old

LOGICALLY VALID SELECTIONS

If p (*vowel/beer*) then q (*odd/20*)

- Select *p (E/beer)*:
  - if ¬ *q*, rule not true
- Don't select *¬p (C/coke)*:
  - no info about rule
- Don't select *q (5/25y.o.)*:
  - neither *p* nor *¬p* leads to problems
- Select *¬q (4/16y.o.)*:
  - if *p* on the other side, rule is not true

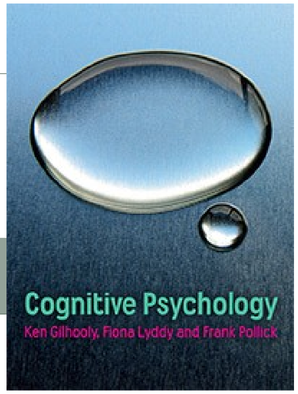
With abstract versions, we are bad (±10% correct).
With some other versions, we are good.

HENCE

- Some problems are easy, some are difficult.
- Content has a big effect.
- Hard to explain by mental logic:
  - Why difference in translation in abstract terms?

1. DEDUCTION

MENTAL MODELS



MENTAL MODEL THEORY

- Probably the most popular theory of reasoning (Johnson-Laird):
  - Discourse comprehension
  - Relational reasoning
  - Conditional reasoning
  - …
  - Also syllogistic reasoning
- Widely applicable + good account of data

BASIC MMT

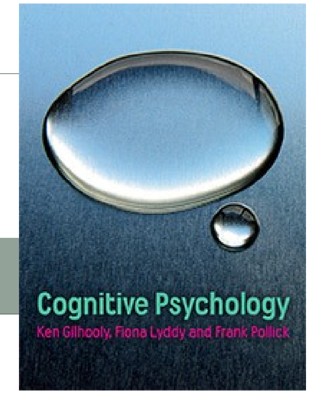
- The mind:
  - ≠ a logical or a probabilistic device,
  - = device that makes mental simulations.
- If humans reason logically or infer probabilities, they rely on their ability to simulate the world in mental models.
- The application of simulation to reasoning is based on mental models of the possibilities to which the premises refer.
- A valid deduction has a conclusion that holds in all these models.

BASIC MMT

- Mental models are representions of what is common to a distinct set of possibilities
- Analogy:
  - Models of architects
- The more (or more complex) models that have to be constructed, the harder a problem is:
  - the more likely it is that reasoners err
  - the longer it takes before they reach a conclusion

1. DEDUCTION

SPATIAL REASONING



WHAT IS SPATIAL REASONING?

- Reasoning that hinges on the spatial relations between events.
- Use of "*to the left of*", "*to the right of*", …
- Example:

  The fork was on the left of the spoon.

  The knife was on the right of the spoon.
- Different theories try to explain it.

MENTAL MODEL ACCOUNT

A is to the left of B.

B is to the left of C.

D is in front of A.

E is in front of C.

<u>D & E?</u>

MENTAL MODEL ACCOUNT

A is to the left of B.
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<u>D & E?</u>
A B

MENTAL MODEL ACCOUNT

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A B

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B is to the left of C.
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E is in front of C.
<u>D & E?</u>
A B C

MENTAL MODEL ACCOUNT

A is to the left of B.
B is to the left of C.
D is in front of A.
E is in front of C.
D & E?
A B C
D

MENTAL MODEL ACCOUNT

A is to the left of B.
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D E

It captures what is common to all the different ways in which the possibility may occur.

MENTAL MODEL ACCOUNT

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D & E?

A B C
D E

One-model problem

MENTAL MODEL ACCOUNT

A is to the left of B.
B is to the left of C.
D is in front of A.
E is in front of C.
<u>D & E?</u>
A B C
D   E
One-model problem

A is to the left of B.
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<u>D & E?</u>

MENTAL MODEL ACCOUNT

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  E D

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D & E?
A B C
D E
One-model problem

A is to the left of B.
C is to the left of B.
D is in front of B.
E is in front of C.
D & E?
A C B   C A B
E D   E D
Multiple-model problem

MENTAL MODEL ACCOUNT

A is to the left of B. ***easier than*** A is to the left of B.
B is to the left of C. C is to the left of B.
D is in front of A. D is in front of B.
E is in front of C. E is in front of C.
D & E? D & E?

A B C
D E
One-model problem

A C B / E D   C A B / E D
Multiple-model problem

MENTAL MODEL ACCOUNT

A is to the left of B.
C is to the left of B.
D is in front of B.
E is in front of C.
D & E?

A C B / E D   C A B / E D
Multiple-model problem

MENTAL MODEL ACCOUNT

A is to the left of B.
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D & E?

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A C B   C A B
E D   E   D
Multiple-model problem

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A C B   C A B
E D   E   D
Multiple-model problem

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D E

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A C B   C A B
E D   E   D
Multiple-model problem

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D E   E D

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<u>D & E?</u>
A C B   C A B
E D   E   D
Multiple-model problem

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D & E?
A C B   C A B
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Multiple-model problem

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D & E?
A C B   C A B
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E is in front of C.
D & E?
A C B   C A B
E D   E   D
Multiple-model problem

MENTAL MODEL ACCOUNT

A is to the left of B.
C is to the left of B.
D is in front of A.
E is in front of C.
D & E?
A C B   C A B
D E   E D
No-valid-conclusion

A is to the left of B.
C is to the left of B.
D is in front of B.
E is in front of C.
D & E?
A C B   C A B
E D   E   D
Multiple-model problem

MENTAL MODEL ACCOUNT

A is to the left of B.
C is to the left of B.
D is in front of A.
E is in front of C.
D & E?
A C B   C A B
D E   E D
No-valid-conclusion

harder than

A is to the left of B.
C is to the left of B.
D is in front of B.
E is in front of C.
D & E?
A C B   C A B
E D   E   D
Multiple-model problem

MENTAL LOGIC ACCOUNT

- Each premise has its own representation and is independently stored in memory
- Next inference rules are applied to them, until the required relation is derived:
  - E.g., Left(x,y) & Left(y,z) => Left(x,z)

MENTAL LOGIC ACCOUNT

MENTAL LOGIC ACCOUNT

- 1M-problem

  A left of B
  B left of C
  D in front of A
  E in front of C

  A B C
  D E

MM-problem

  A left of B
  A left of C
  D in front of A
  E in front of C

  A B C
  D E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

  A left of B
  B left of C
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  E in front of C

  A B C
  D E

MM-problem

  A left of B
  A left of C
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  E in front of C

  A B C
  D E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

  A left of B
  B left of C
  D in front of A
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  A B C
  D   E

MM-problem

  A left of B
  A left of C
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  A B C
  D   E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

  A left of B
  B left of C
  D in front of A
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  A B C
  D   E

MM-problem

  A left of B
  A left of C
  D in front of A
  E in front of C

  A B C
  D   E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

  A left of B
  B left of C
  D in front of A
  E in front of C

  A B C
  D   E

MM-problem

  A left of B
  A left of C
  D in front of A
  E in front of C

  A B C
  D   E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

  A left of B
  B left of C
  D in front of A
  E in front of C

  A B C
  D   E

MM-problem

  A left of B
  A left of C
  D in front of A
  E in front of C

  A B C
  D   E

  A C B
  D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

A left of B
B left of C
D in front of A
E in front of C

A B C
D E

MM-problem

A left of B
A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

A left of B
B left of C
D in front of A
E in front of C

A B C
D E

MM-problem

A left of B
A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

A left of B
B left of C
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A B C
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MM-problem

A left of B
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MENTAL LOGIC ACCOUNT

- 1M-problem

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MM-problem

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MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

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MM-problem

A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- 1M-problem

Transitivity-rule is used once

A left of B
B left of C
D in front of A
E in front of C

A B C
D E

MM-problem

Transitivity-rule not used

A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- 1M-problem ***<u>harder than</u>*** MM-problem

Transitivity-rule is used once

A left of B
B left of C
D in front of A
E in front of C

A B C
D E

Transitivity-rule not used

A left of B
A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- NVC-problem

A left of B
A left of C
D in front of B
E in front of C

A B C
D E

A C B
E D

MM-problem

A left of B
A left of C
D in front of A
E in front of C

A B C
D E

A C B
D E

MENTAL LOGIC ACCOUNT

- NVC-problem
  A left of B
  A left of C
  D in front of B
  E in front of C

  A B C   A C B
  D E   E D

MM-problem
A left of B
A left of C
D in front of A
E in front of C

A B C   A C B
D E   D E

MENTAL LOGIC ACCOUNT

- NVC-problem
  A left of B
  A left of C
  D in front of B
  E in front of C

  A B C   A C B
  D E   E D

MM-problem
A left of B
A left of C
D in front of A
E in front of C

A B C   A C B
D E   D E

MENTAL LOGIC ACCOUNT

- NVC-problem
  A left of B
  A left of C
  D in front of B
  E in front of C

  A B C   A C B
  D E   E D

MM-problem
A left of B
A left of C
D in front of A
E in front of C

A B C   A C B
D E   D E

MENTAL LOGIC ACCOUNT

- NVC-problem ***harder than*** MM-problem
  A left of B
  A left of C
  D in front of B
  E in front of C

  A B C   A C B
  D E   E D

A left of B
A left of C
D in front of A
E in front of C

A B C   A C B
D E   D E

EXPERIMENTAL DATA

- Byrne & Johnson-Laird, 1989; Carreiras & Santamaria, 1997; Ragni, & Knauff, 2013; Roberts, 2000; Schaeken, Johnson-Laird & d'Ydewalle, 1996a; 1996b; Schaeken, Girotto & Johnson-Laird, 1998; Schaeken & Johnson-Laird, 2000; Schaeken, Van der Henst, & Schroyens, 2007; Van der Henst 1999, 2002; Van der Henst & Schaeken, 2005; …
- Predictions MMT:
  - %: 1M > MM > NVC
  - rt.: 1M < MM < NVC

  Predictons MLT:

  MM >1M>NVC

  MM <1M<NVC
- In general:
  - %: 1M ≥ MM ≥ NVC
  - rt.: 1M ≤ MM ≤ NVC

COMPUTATIONAL ACCOUNTS OF SPATIAL REASONING

- Until now description in words, however many attempts for computational models, many in line with mental models theory
- Hummel and Holyoak (2001):
  - a mental array module (MAM) that maps pairwise relations between objects onto locations in a spatial array.
  - Part of LISA a more general model of human relational inference (Hummel & Holyoak, 2003, 2005), a connectionist model with some symbolic ingredients
- Bara et al. (2001):
  - UNICORE, relies on the placement of tokens in a spatial array and a scanning process that realizes the inference
  - one large algorithm difficult to be split into its individual subprocesses
- See also Baguly and Payne (2000); Johnson-Laird & Goodwin (2007); Schlieder& Berendt, (1998)
- We discuss in a bit more detail PRISM by Ragni & Knauff (2013)

SOME ASSUMPTIONS OF PRISM

- Indeterminate problems:
  - People are likely to construct just a single, simple, and typical model, even when a description is compatible with several alternative models
  - = preferred mental model.
- Simple principle for preferred mental models of spatial descriptions:
  - new objects are added to a model without disturbing the arrangement of those tokens already represented in the model.
  - `= First-free-fit-strategy
  - *A left of B; C left of B*

	A	B	

C	A	B	

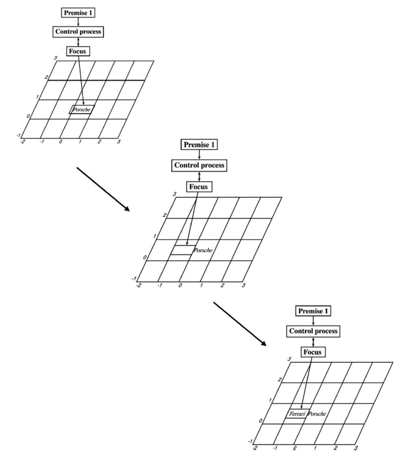
SOME ASSUMPTIONS OF PRISM

- People search for alternative interpretations of the premises only if this is explicitly required:
  - If search for alternative models is required, it always starts with the preferred model.
  - Alternative models are constructed by local transformations, and the process follows the principle of minimal changes.
- The number of necessary focus operations is a measure for the difficulty of an inference rather than the number of possible models.
  - It's more specific and more precise than the number of models

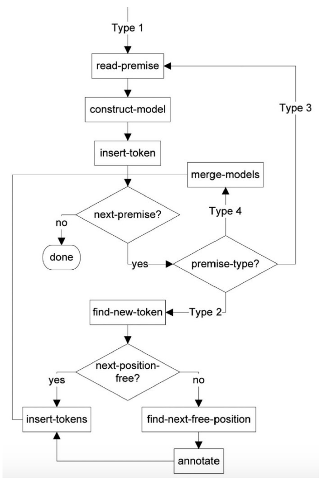
PRISM

- Four types of premises:
  - Type 1—Initial premise: starting point of model construction
  - Type 2—One-new-token premise:
    - has 2 tokens, but one already mentioned; two versions:
      - Type 2d: there is a fixed place for the new token (determinate)
      - Type 2i: there is more than one possibility for the new token (indeterminate)
  - Type 3—Two-new-tokens premise:
    - None of the tokens mentioned in previous premises
    - Can happen with discontinuous presentations
  - Type 4—Connecting-submodels premise:
    - with a token that connects two partial models

PRISM AT WORK: THE SPATIAL ARRAY FOR *"THE PORSCHE IS TO THE RIGHT OF THE FERRARI."*



PRISM AT WORK



PRISM AT WORK: 1M-PROBLEM

Premise	Mental model	Operation of the focus	Accumulated number of focus operations
		Determinate problem with valid conclusion (Type 1 problem)	
Model construction			
A is right of B.	A	Insertion of object A	1
	_ A	Moves to the left	2
	B A	Insertion of object B	3
C is left of B.	_ B A	Moves one step left	4
	C B A	Inserts object C	5
D is in front of C.	C B A	Move one step in front of object C	6
	_ C B A D	Insert object D	7
E is in front of B.	C B A D	Search object B	8
	C B A D	Move one step in front of object B	9
	C B A D _	Move one step in front of object B	10
	C B A D E	Insert object E	11
Model inspection			
D left of E? Yes or No?	C B A D E	Focus is at E after model construction	11
	C B A D E	Move left (according to the relation in the conclusion)	12
PRISM output		YES, D is to the left of E. (which is logically correct)	12

PRISM AT WORK: MM-PROBLEM

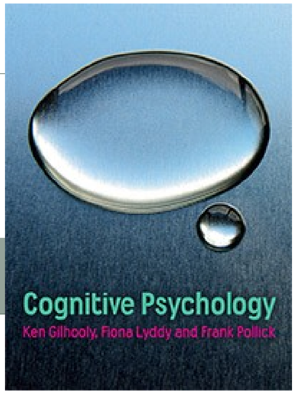
Indeterminate problem with valid conclusion (Type 2 problem)

Model construction			
B is right of A.	B	Insertion of object B	1
	_ B	Move one step left	2
	A B	Insertion of object A	3
C is left of B.	A B	Search reference object B	4
	A B	Move one step left (ff-principle, not empty)	5
	_ A B	Move one step further to the left	6
	C_B A B	Insert object C (fff-principle) with annotation B (somewhere left of B)	7
D is in front of C.	C_B A B _	Move one step in front of object C	8
	C_B A B D	Insert object D	9
E is in front of B.	C_B A B D	Search object B	10
	C_B A B D	Search object B	11
	C_B A B D	Move one step in front of object B	12
	C_B A B D _	Move one step in front of object B	13
	C_B A B D E	Insert object E— preferred model	14
Model inspection			
D left of E? Yes or No?	C_B A B D E	Focus is still at E after model construction	14
	C_B A B D E	Move left (according to the relation in the conclusion)	15
PRISM output		YES, D is to the left of E. (which is logically correct)	15

PRISM AND OTHER MENTAL-MODEL INSPIRED PROGRAMS

- https://www.modeltheory.org/models/

SYLLOGISMS



SYLLOGISMS

SYLLOGISMS

Problem 1:
All A's are B's.
All B's are C's.
What follows?

Problem 2:
None of the A's are B's.
All B's are C's.
What follows?

SYLLOGISMS

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SYLLOGISMS

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Problem 2:
None of the A's are B's.
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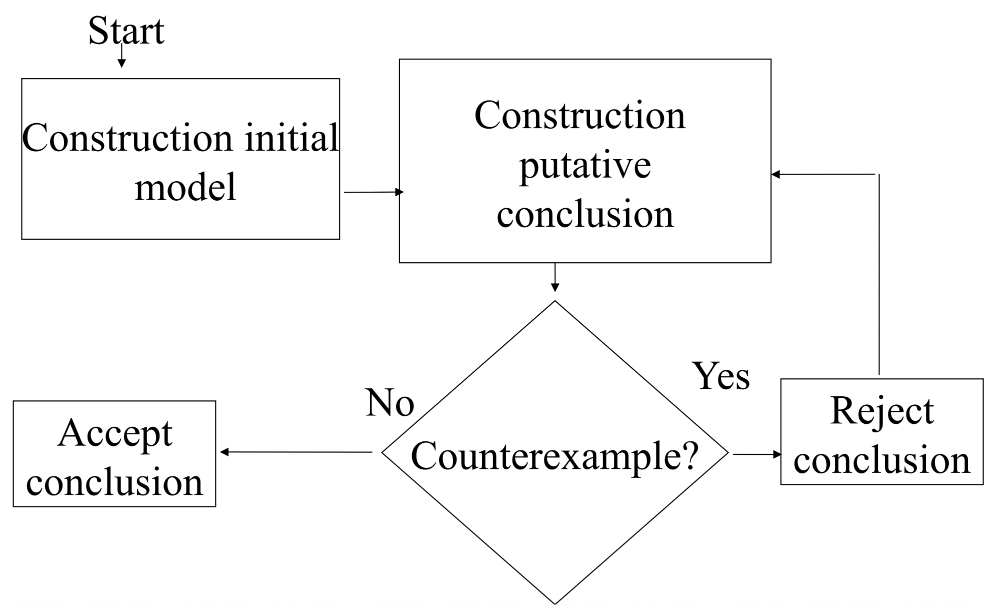
What follows?
Some of the C's are not A's.

SYLLOGISMS



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model 1	model 2	model 3
A	A	A = C
A	A = C	A = C
B = C	B = C	B = C
B = C	B = C	B = C
other C	other C	other C
None A = C None C = A	Sommige A ≠ C	Sommige C ≠ A

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None C = A		

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B = C	B = C	B = C
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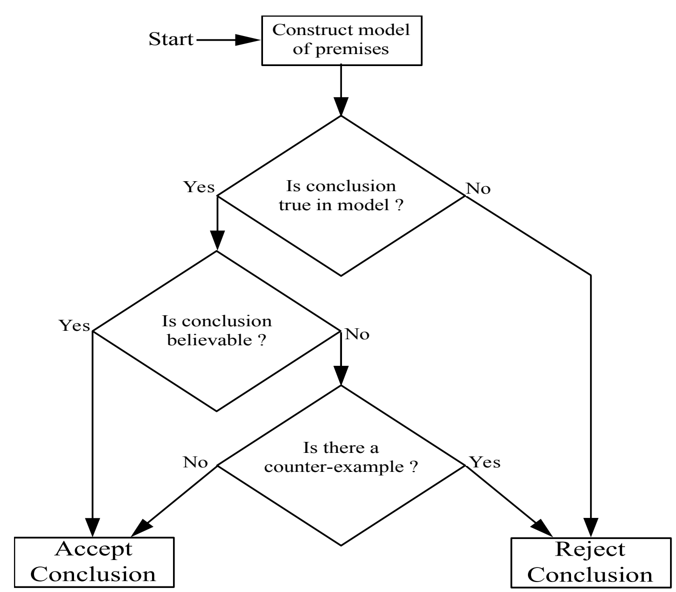
RESULTS

- The more models, the more difficult:
  - More mistakes
  - More time
- Conclusion from left to right is easier
- Content effects:
  - Model construction
  - Model validation!!!!

MM AND CONTENT WHEN EVALUATING A CONCLUSION



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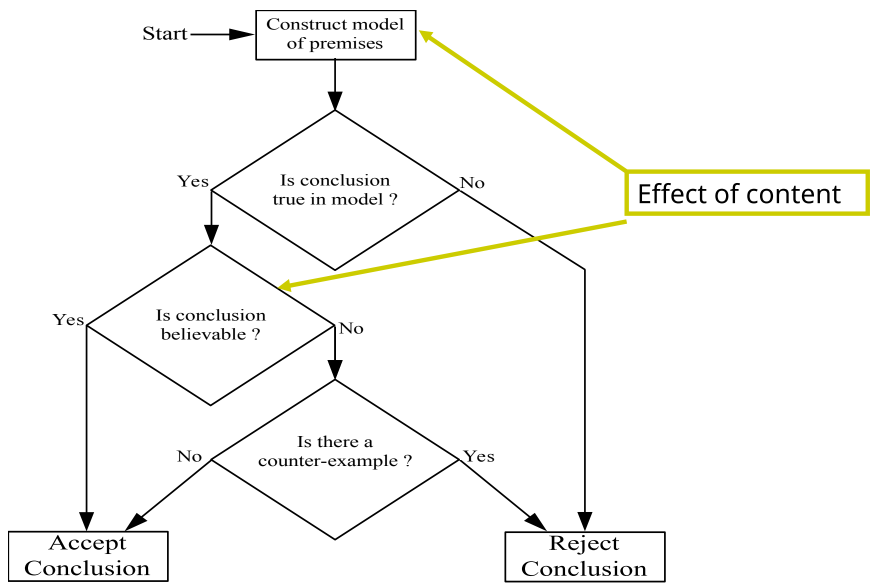

MM AND CONTENT WHEN GENERATING A CONCLUSION